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Roll No

EC-8012 (GS)
B.E. VIII Semester Examination, June 2020
Grading System (GS)
Microwave Circuits
(Elective - II)
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Explain (i) single stub matching (ii) Double stub matching. Enumerate the advantages and disadvantages of each of these methods.
2. Design a single section quarter wave matching transformer to match 350 ohms load to a 100 ohms line. What is the percentage of bandwidth of this transformer for $SWR \leq 2$?
3. What is effective dielectric constant for a microstrip line? Write all necessary equations for a strip line using capacitance model. Derive an approximate Quasi-static solution for the microstrip line.
4. A load impedance $Z_L = 100 + j80 \Omega$ is to be matched to a 75Ω line using a single shunt-stub tuner. Find the two solutions using open-circuited stubs.
5. Which mode of propagation is supported by a strip line? If the loss tangent is 0.001 for a stripline operating at 12GHz with the relative permittivity of the dielectric material being used equal to 2.6, find the conductor loss.
6. a) What is quarter wave transformer? Give its characteristics and applications.
b) Why impedance matching is done? Define and explain impedance mismatch factor.
7. A microwave transistor has the following S parameters: $S_{11} = 0.8 \angle -90^\circ$, $S_{21} = 5.1 \angle 80^\circ$, $S_{12} = 0.3 \angle 70^\circ$, $S_{22} = 0.62 \angle -40^\circ$. Determine the stability, and plot the stability circles if the device is potentially unstable.
8. Write short notes on any two of the following.
 - a) Smith chart
 - b) Binomial and chebyshev transformer
 - c) Power amplifiers
 - d) Microstrip half-wave filter
 - e) Narrow band and wideband microwave filters.
